Paper Review | Embodied AI Evaluation

Frontier VLM Agents: Ready to Be Robot Generalists?

Are Frontier VLM Agents Ready to Be Robot Generalists? An Empirical Study with the Embodied Agent Arena

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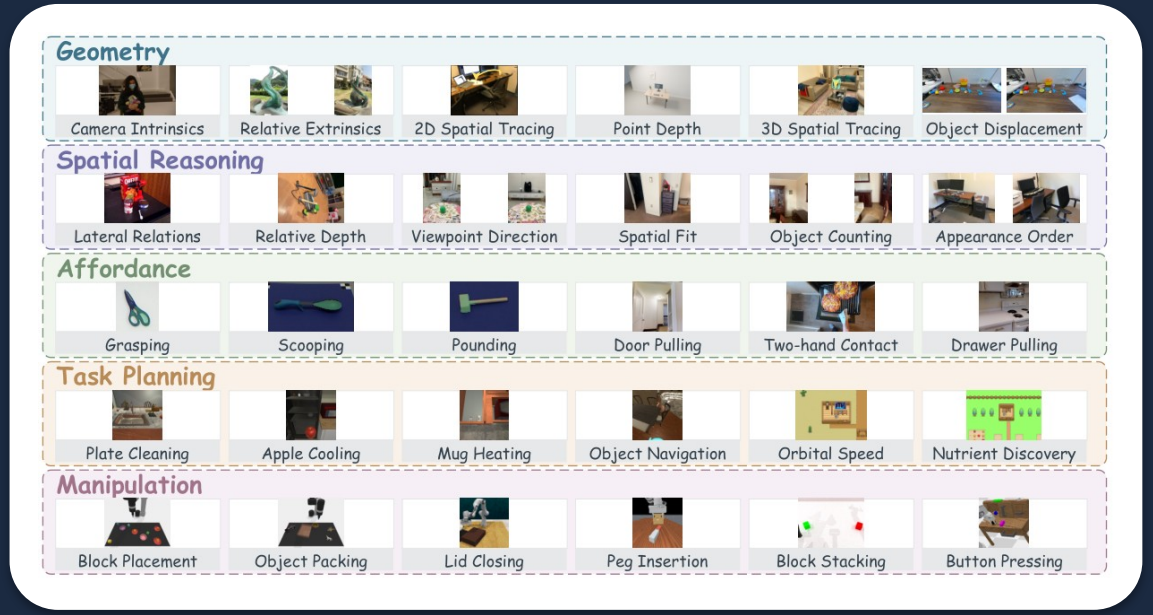


Source: Fig.1, five capability domains × six task types

TL;DR

Not yet: failures often occur after intermediate steps succeed

1,000
Cases | 32 existing benchmarks + new GeoProbe (168 cases), five domains

10
frontier VLM agents | 7 proprietary services + 3 open-weight VLMs

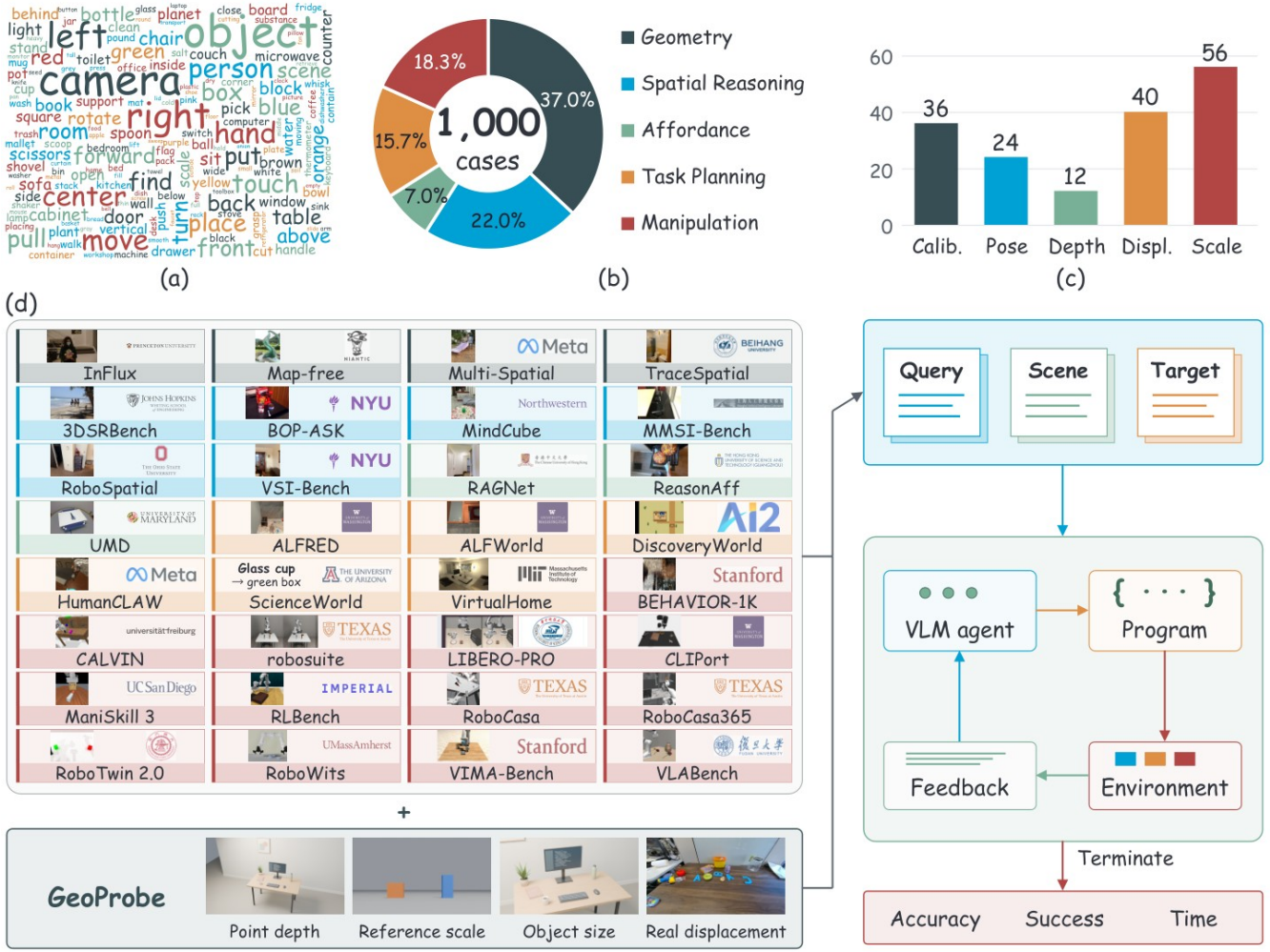
47.8%
top manipulation success (Opus 5.5, 182 tasks); best planning 80.3% (Astra)

3 / 65
CALVIN error episodes: code resumed in 62, yet only 3 finished the task

Key Findings

- **Proprietary models lead; open models lag in execution:** open VLMs reach 48–52% in spatial reasoning but only 19–31% in planning and 10–21% in manipulation.
- **Averages hide distinct errors:** scale bias vs. shape distortion; correct region box vs. correct action point; runnable code vs. achieved object state.
- **Extra compute is not reliably helpful:** code fixes restore execution but not completion; self-review turns right answers wrong; failure tokens concentrate in long-horizon body control.

Arena: 1,000 cases × 5 domains, native interfaces + unified loop



Composition

- Geometry 370 • Spatial 220 • Affordance 70
- Planning 157 • Manipulation 183
- 660 fixed-input items + 340 interactive episodes
- GeoProbe: controlled Blender variables separating camera motion, object motion and scale reference

Harness Principles

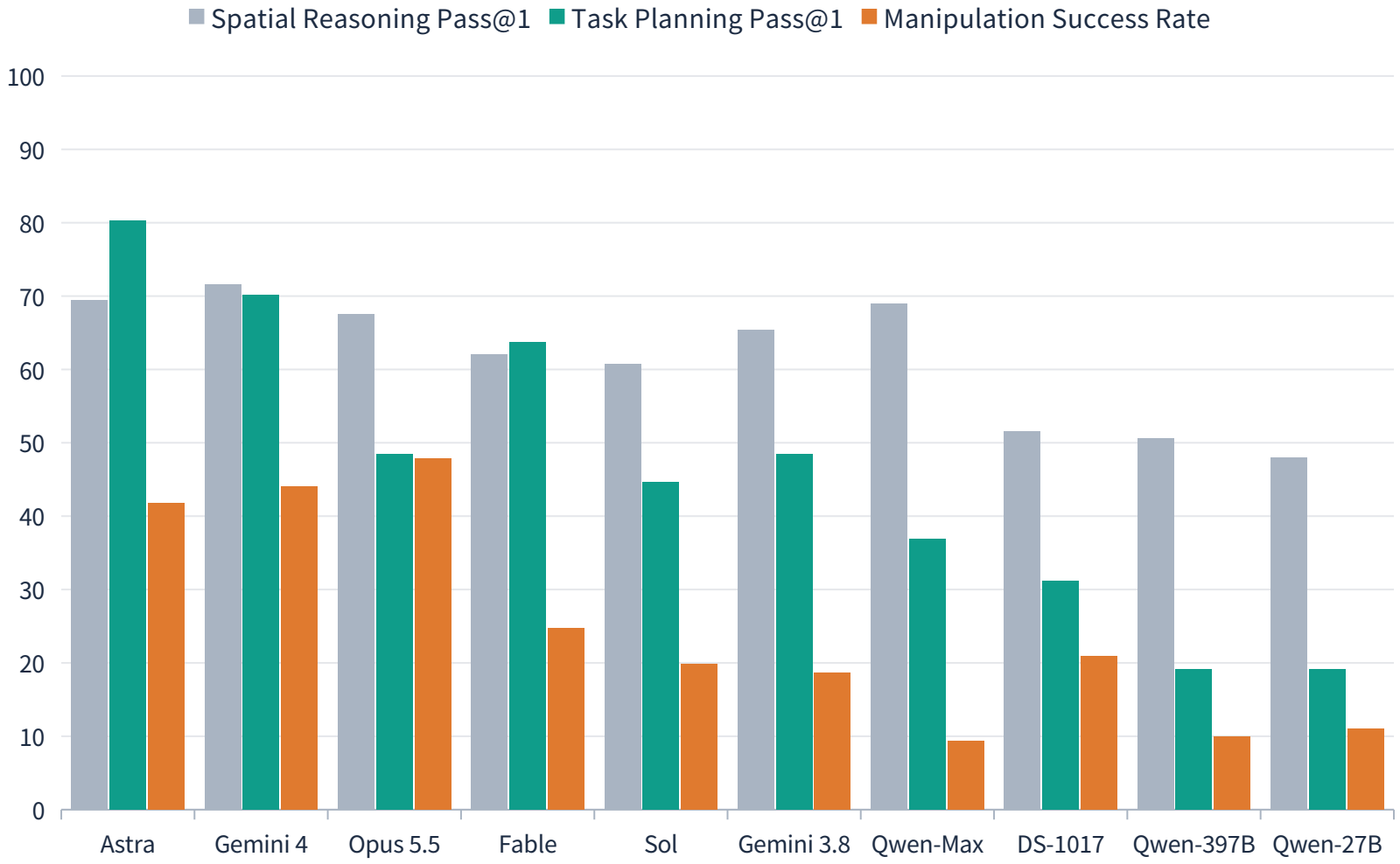
Keep native interfaces: each source's observations, actions and success criteria; no added perception or control modules

Model makes decisions: persistent Python runtime; images and variables persist across turns

Termination ≠ success: an independent evaluator checks the final state; all predicates of compound goals must hold

Main Results

Strong perception ≠ execution: similar spatial scores, big action gaps



36.9 vs 80.3

Planning success of Qwen-Max vs. Astra; their spatial reasoning is nearly equal (68.9 vs 69.4)

47.8%

Opus 5.5 ranks first in manipulation, yet trails Astra and Fable in planning (48.4%)

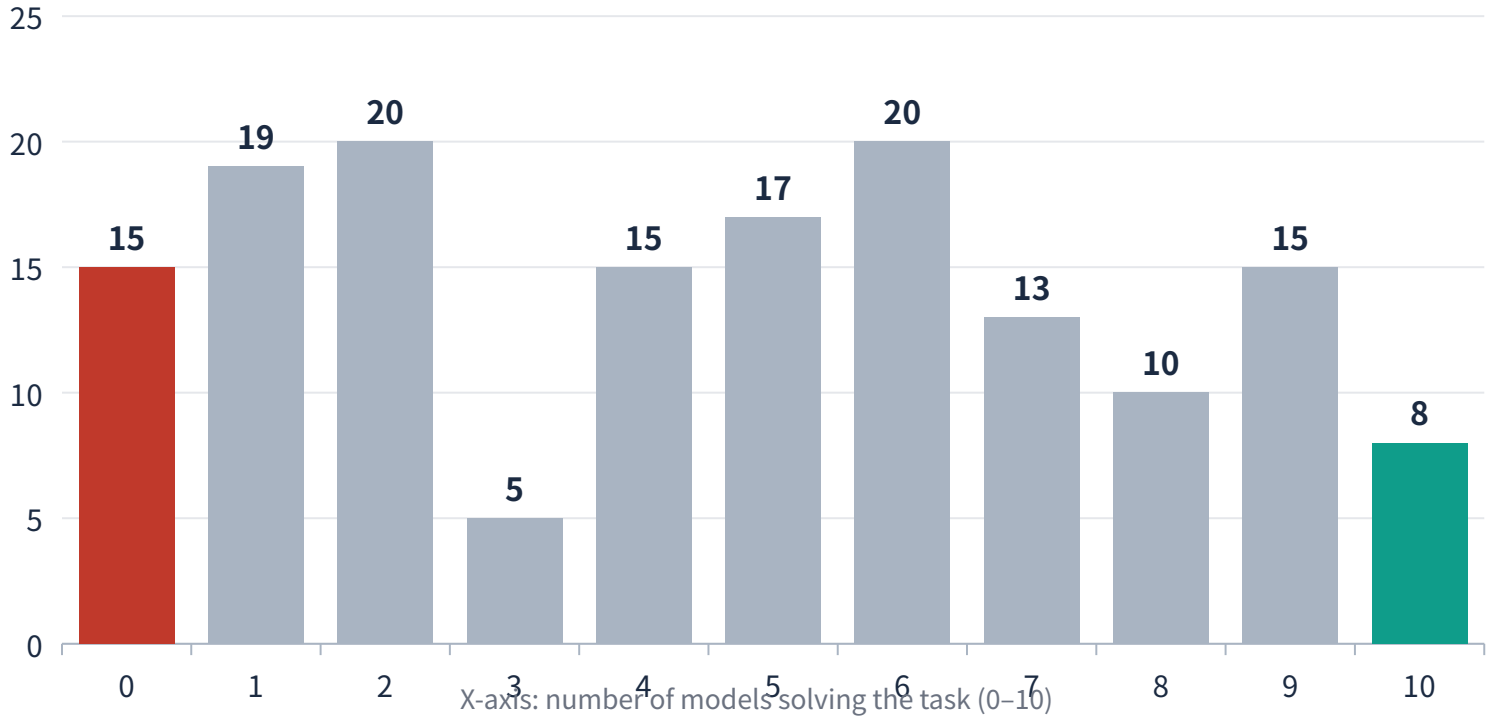
10–21%

Manipulation success of the three open VLMs; spatial reasoning is 48–52%

Task Planning Overlap Analysis

Limited complementarity: 10 models add only 16 tasks; 15 unsolved

157 planning tasks: task count by number of models solving each task



- Only 8 tasks are solved by all 10 models; 7 of them are symbolic VirtualHome tasks
- 39 tasks are solved by only 1–2 models; 20 of them are ALFRED/ALFWorld visual household tasks

Diminishing Coverage Gains

126 Astra alone

137 + Gemini 4 (11 more tasks)

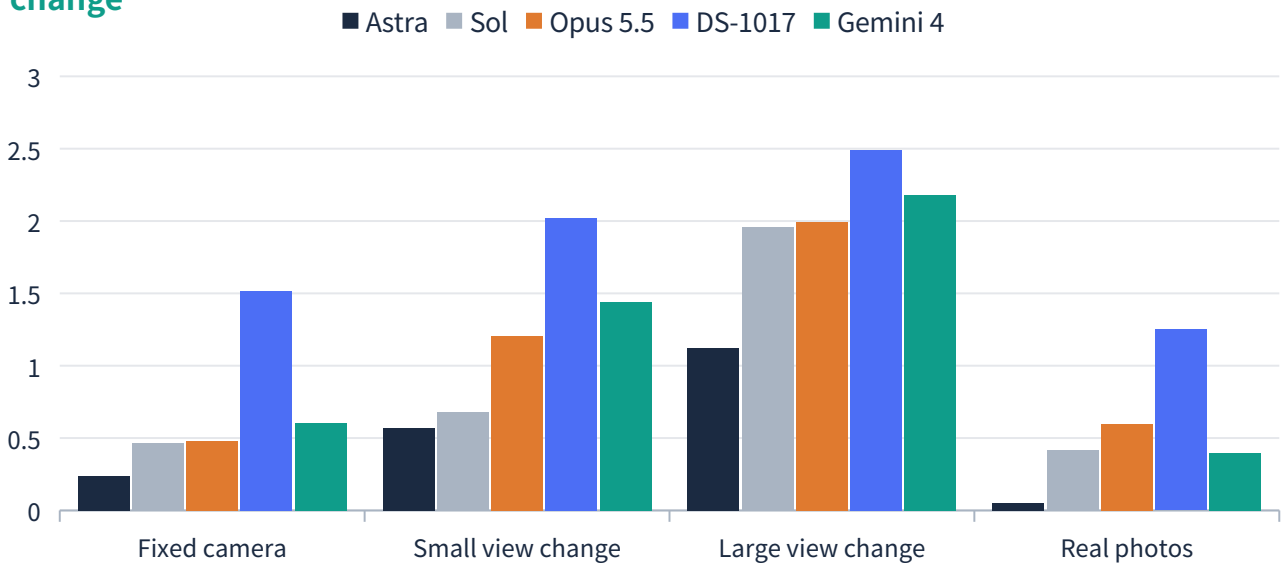
142 All 10 models (other 8 add only 5)

The 15 shared failures are heterogeneous: 8 "sit" tasks, 4 ALFRED/ALFWorld, 2 ScienceWorld, 1 TV task, each needing a different fix.

Diagnosis ① Geometry

Geometry: similar errors may stem from scale bias or shape distortion

GeoProbe displacement MAE (cm, lower is better): error grows with camera change



TraceSpatial 3D trajectory joint-constraint success (%)

0.299→0.065

Gemini 4 depth AbsRel drops sharply after oracle scale alignment; Qwen-397B 0.312→0.203, residual is mostly shape error

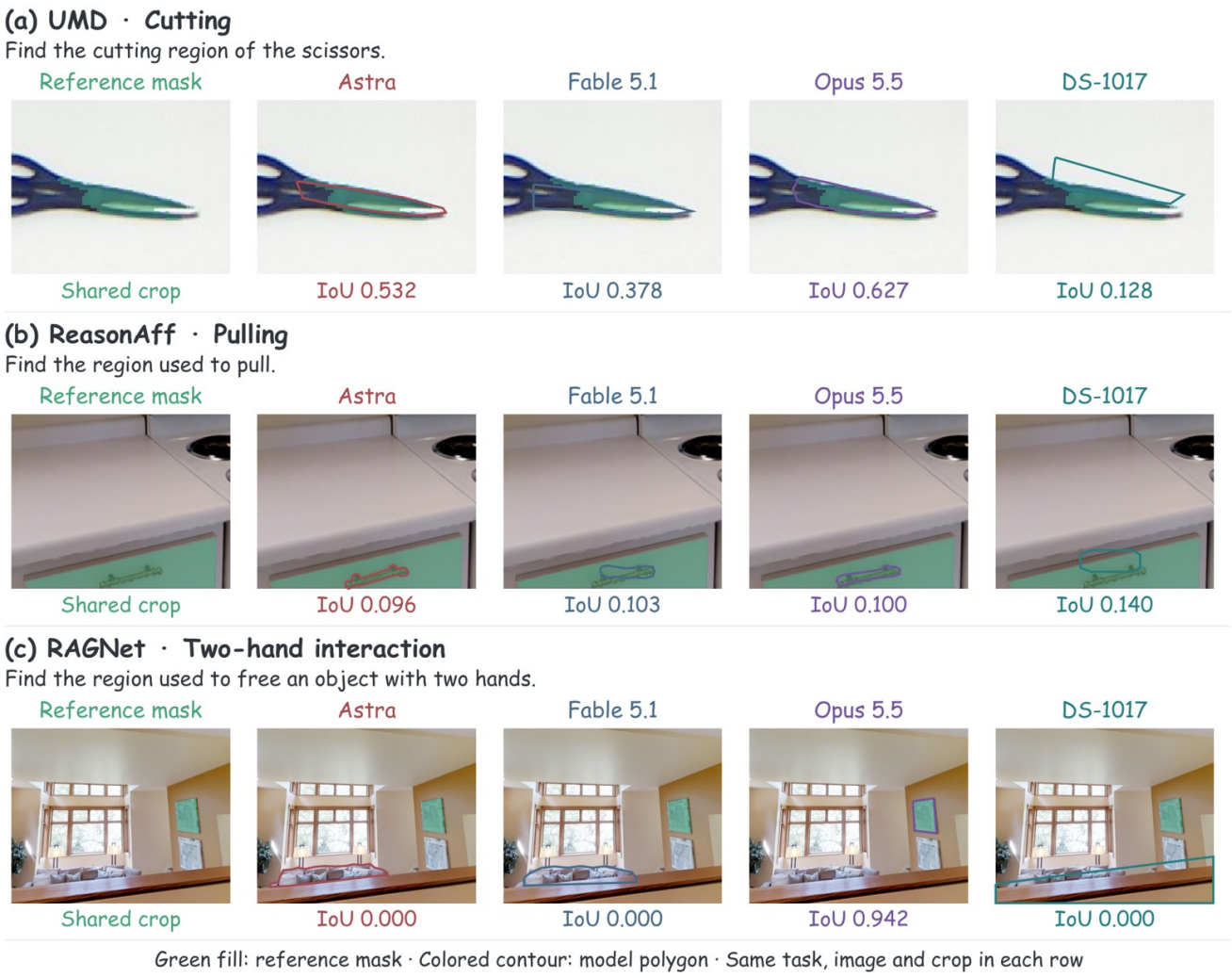
0 → 26 / 37

Astra / Gemini 4 miss none of 50 motion samples with a fixed camera, but miss 26 / 37 under large view changes; Qwen-27B misses all in every condition

21 / 35

Of Astra's 35 trajectories with both endpoints hit in the image, 21 still violate 3D start/end constraints; Gemini 4 has 10 more failing only due to collision

Affordance: right object, region coverage and action point differ



96 vs 18

Qwen-397B: of 350 predictions, 96 pass the point criterion; only 18 pass both point and box

IoU 0.784, point outside

Gemini 4: 16 box passes with points outside the functional region (scissors grasp, 4 of 5 runs)

0.942 vs 0

RAGNet bimanual task: Opus picks the right target; Astra/Fable/DeepSeek box other objects

9 / 10

models get higher IoU with polygons than boxes; interaction scenes stay far harder than tools

arXiv 2610.00854 | Source: Fig.A7 (polygon outputs, green = reference mask); Data: Table A46, Fig.6, Section 4.3

Planning: executing an action ≠ reaching the required state

Case A | ALFRED: wash soap, put in bathtub

Wrong receptacle binding

DeepSeek	Wash soap → place in Bathtub (outer fixture)	1/3
Gemini 3.8	Check parent → pick up again → place in BathtubBasin	3/3
Qwen-Max	Check parent → rewash → rebind receptacle to Basin	3/3

Feedback helps only if it changes the target binding rather than repeating the original action.

Case B | HumanCLAW: find the sofa and sit

Mistaking "attempted interaction" for the goal state

7 / 8

Gemini 4 got within 20 cm and issued sit

0 / 8

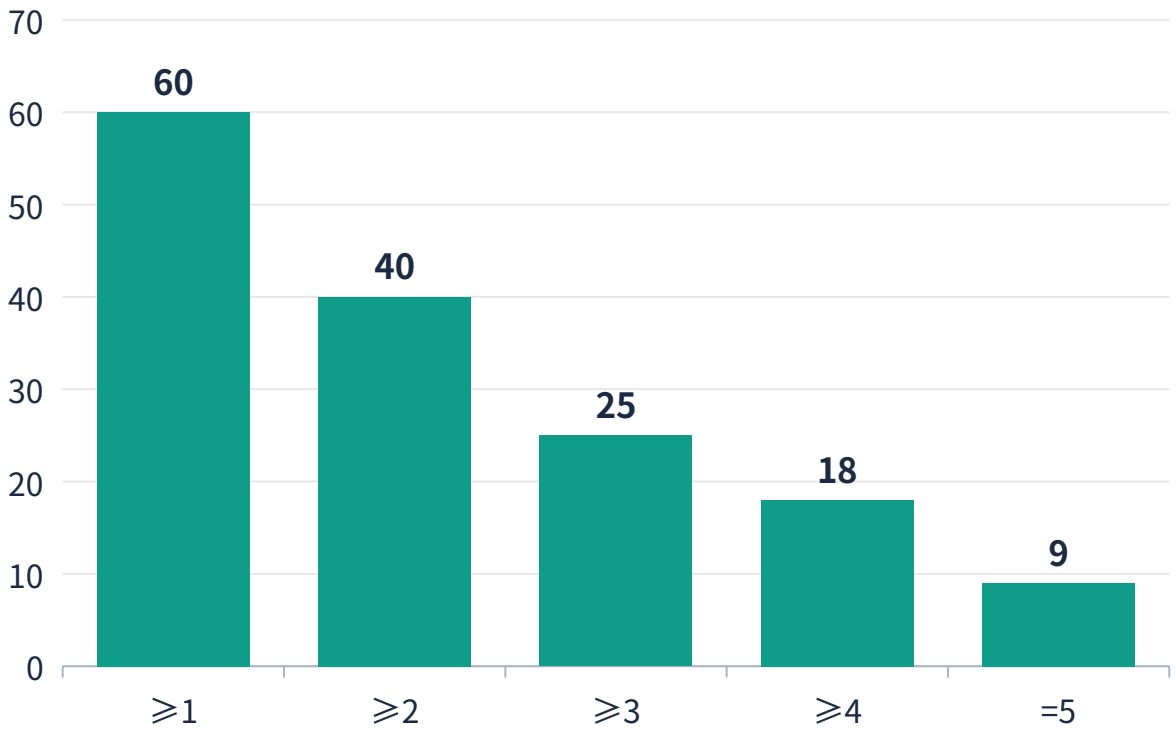
Sit-task successes across all 10 models

- Same sofa case: Gemini 4 and DeepSeek issued sit, claimed to be "seated", and stopped at turns 44 and 70; both failed the private verifier
- Approaching the target + issuing a command ≠ establishing a body–object relation; these are stopping-decision errors

Diagnosis ④ Manipulation

Manipulation: starting is easy, finishing hard; converging ≠ grasping

CALVIN: episodes reaching ⩾k consecutive goals (of 100)



Of 60 episodes completing the first goal, only 9 finish the full chain; Gemini 4: 10→4, Astra: 8→3.

State Issues from Paired Trajectories

PlaceSphere

DeepSeek's gripper came within 2.8 mm of the hover target, yet 4 checks returned is_grasped = false and the ball was pushed away; Opus verified grasp → release → on box and at rest: success.

CALVIN 0526

Opus grasped at turn 11 and stacked at turn 13, 18 turns total; DeepSeek re-grasped repeatedly in turns 17–27 (cube height fell to ~2 cm), 37 turns total.

Precision Insertion

Opus completes PickCube / PlaceSphere / StackCube but fails PegInsertionSide and PlugCharger (requiring 5 mm, 0.2 rad).

Role of Feedback

Compute ≠ progress: fixed code ≠ done; review adds errors

CALVIN: what happens after execution errors

Qwen-Max: 103 of 200 turns had execution errors (95 syntax errors); non-compliant interface use wastes interactions.

Fixed-input multi-turn review (46-case shared subset)

Qwen-27B self-review	**+1 fixed**	**−2 broken**
Qwen-27B crop review	**+2 fixed**	**−4 broken**

Both reviews changed a correct "back" to "left" on 3DSRBench.

0.034→0.152

Self-review fixes all 6 out-of-bounds boxes; mean IoU rises

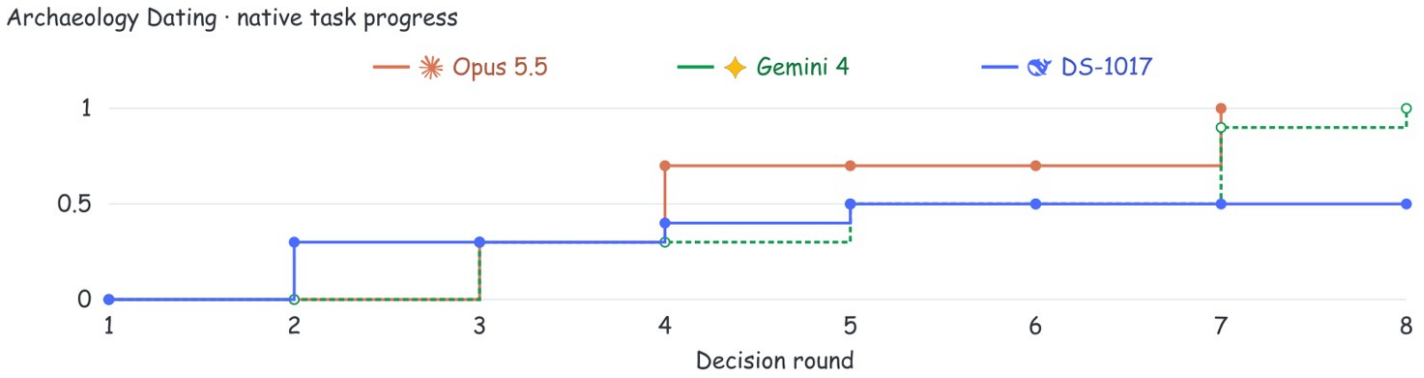
0 / 9

but still no box meets the IoU ⩾ 0.5 criterion

Representation validity is fixed, but functional-region error remains: measure semantic fixes, coordinate validity and task acceptance separately.

What Successful Recovery Looks Like

Good recovery fixes preconditions so feedback advances the rest



DiscoveryWorld archaeology dating: native task score per turn (step line = latest score)

Opus 5.5

Errors in first two turns before any environment action → get tools → dig three sites → read artifact ages → flag oldest site; 1.0 at turn 7

Gemini 4

Error at turn 2, progress resumes at turn 3, done at turn 8

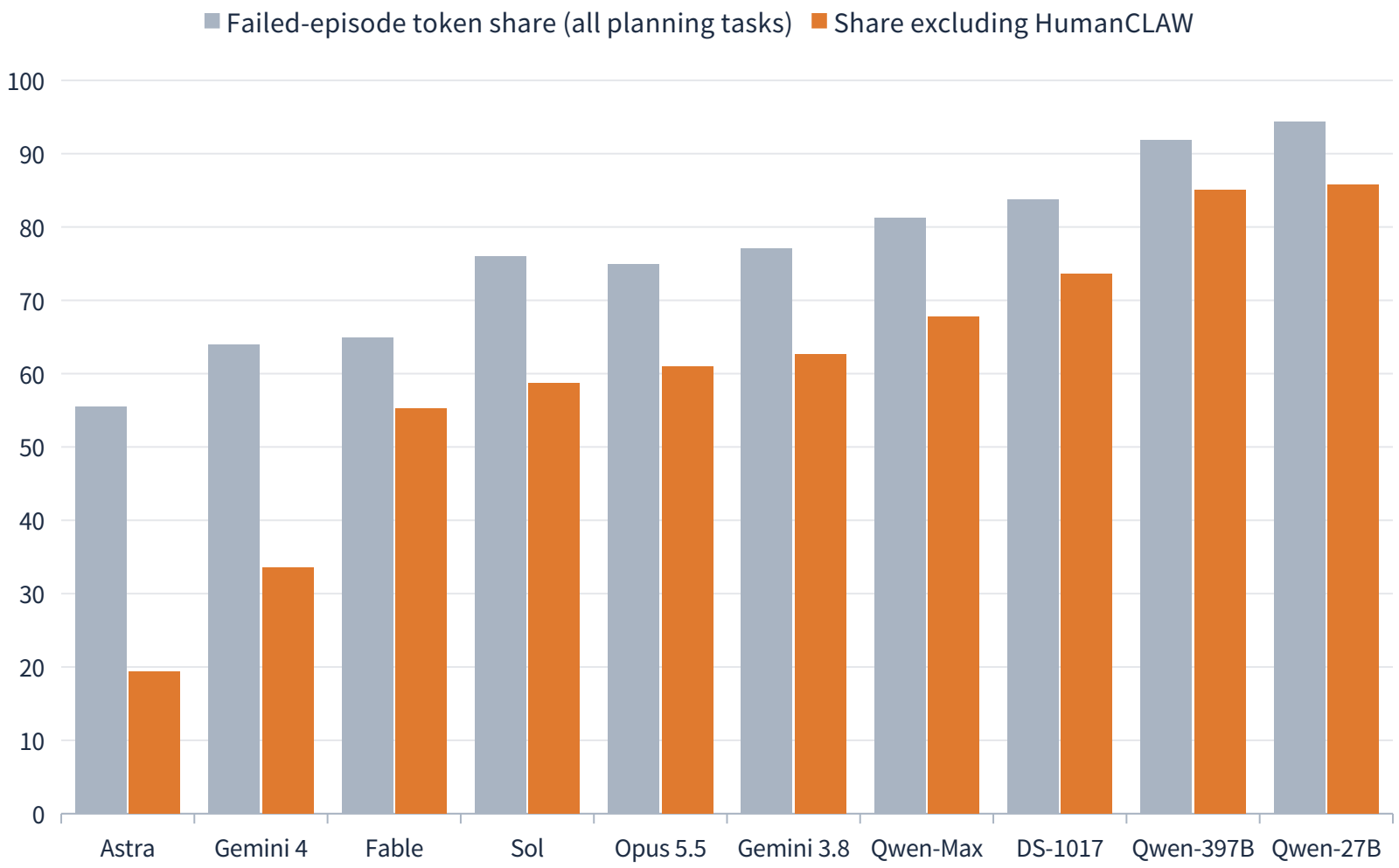
DeepSeek

Reaches 0.5 at turn 5; last three turns are code errors with no new environment steps: stalled

Key: each fix yields evidence for the next decision

Resource Cost

Top planner's failure tokens cluster in long-horizon control



81.7%

Share of Astra's failure tokens from HumanCLAW (13/157 tasks; budget 101 turns / 2M tokens)

55.5→19.4%

Astra's failure-token share excluding HumanCLAW; Gemini 4: 64.0→33.5%

98–99%

Prompt-token share in HumanCLAW trajectories: cost is mainly reprocessing inputs

Contributions and Limitations

Contributions

Evaluation centered on full task requirements: jointly measures prediction accuracy, functional grounding, subgoal completion and final success

Unified arena + GeoProbe: keeps native interfaces and criteria of 32 sources; controlled rendering isolates camera, motion and scale

Fine-grained failure diagnosis: point/box cross-tabs, oracle scale alignment, sequence-prefix survival, nested error→recovery→success counts, paired trajectories

Actionable insights: what recovery should change: target binding, precondition state, or an invalid completion claim

Limitations / Caveats

Small samples: one run per planning/manipulation task; CALVIN 10 sequences, HumanCLAW 13 cases; many conclusions rest on single paired trajectories

Inconsistent execution settings: in CALVIN, 3 later-added models had 40 turns, 1.28M tokens and per-turn RGB; the first 7 had 16–20 turns, 640k tokens

Native capability only: a minimal harness does not reflect the ceiling with dedicated perception/control or VLA policies

Small review study: 46 cases, 2 models; some endpoints are not immutable snapshots, hindering reproducibility

Take-away

Frontier VLMs already show decent spatial perception; the real gaps are maintaining physical state over long horizons, binding action targets correctly, and judging whether a task is truly complete.

Evaluation

Averages alone mislead; also examine subgoals, final-state conditions and failure costs

Feedback

Effective feedback targets the specific condition blocking progress, not more tokens or repeated review

Systems

Making target binding, state verification and stopping decisions explicit may matter more than better perception